

The separable universal metric space

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THE SEPARABLE UNIVERSAL METRIC SPACE

We shall construct a separable space containing a subset isometric to every separable space.

If $p_1, p_2, \dots, p_n$ are n (not necessarily distinct) points of a metric space, then their distances $a_{ik} = p_i p_k$ satisfy the conditions

$$(1) \quad a_{ii} = 0, \quad a_{ik} = a_{ki} \geq 0, \quad a_{ij} + a_{jk} \geq a_{ik}, \quad (i, j, k = 1, 2, \dots, n)$$

The matrices

$$\alpha_n = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \cdots & \cdots & \cdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix},$$

where $n = 1, 2, \dots$ and the entries a_{ik} satisfy conditions (1), are to be defined as elements of a metric space $\mathcal{U}$ in the following way. First, if

$$\alpha_n = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \cdots & \cdots & \cdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \quad (m \leq n)$$

is a *section* of α_n , we define the distance $\alpha_m \alpha_n$ to be the number a_{mn} ($\alpha_n \alpha_m = a_{nm} = 0$). Here, as in what follows, nonidentical matrices may have distance 0 and thus coincide as elements of $\mathcal{U}$. If $\alpha \beta = 0$, we write $\alpha = \beta$.¹ These distances satisfy the triangle axiom. Furthermore,

$$\alpha_m \alpha_n = \max_{\mu < m, \nu < n} |\alpha_m \alpha_\mu - \alpha_\mu \alpha_n|, |\alpha_m \alpha_\nu - \alpha_\nu \alpha_n|$$

because for $m < n$ the right-hand side is

$$= \max_{\nu < n} |\alpha_m \alpha_\nu - \alpha_\nu \alpha_n| \leq \alpha_m \alpha_n,$$

and the maximum is attained for $\nu = m$.

We therefore now postulate in general that

$$(2) \quad \alpha_m \beta_n = \max_{\mu < m, \nu < n} |\alpha_m \alpha_\mu - \alpha_\mu \beta_n|, |\alpha_m \beta_\nu - \beta_\nu \beta_n|,$$

where α_μ, β_ν run through the sections of α_m, β_n . (If $m = 1$ or $n = 1$, the corresponding terms are omitted. We set $\alpha_1 \beta_1$ equal to 0, since all the matrices satisfy $\alpha_1 = \beta_1 = (0)$.) Formula (2) defines $\alpha_m \beta_n$ for $m + n = s$ once the distances are known for $m + n < s$.

¹Equality of matrices is thus defined differently here from the usual matrix calculus. Moreover, $\alpha \beta$ does not denote a product here.

For example, $\alpha_2\beta_2 = |\alpha_2\alpha_1 - \alpha_1\beta_2| = |a_{12} - b_{12}|$.
 $\alpha_3\beta_2 = \max |\alpha_3\alpha_1 - \alpha_1\beta_2|, |\alpha_3\alpha_2 - \alpha_2\beta_2| = \max |a_{13} - b_{12}|, |a_{23} - b_{12}|$.

The triangle axiom now holds in general,

$$(3) \quad \alpha_m\beta_n + \beta_n\gamma_p \geq \alpha_m\gamma_p.$$

We prove this by induction from $m+n+p < s$ to $m+n+p = s$. (It holds for $m+n+p = 3$, that is, $m=n=p=1$.) We have

$$\alpha_m\gamma_p = \max_{\mu < m, \pi < p} |\alpha_m\alpha_\mu - \alpha_\mu\gamma_p|, |\alpha_m\gamma_\pi - \gamma_\pi\gamma_p|.$$

By symmetry, we may assume that one of the terms indexed by μ attains the maximum. Let $\alpha_m\gamma_p = |\alpha_m\alpha_\mu - \alpha_\mu\gamma_p|$. Then

$$\alpha_m\gamma_p \leq |\alpha_m\alpha_\mu - \alpha_\mu\alpha_n| + |\beta_n\alpha_\mu - \alpha_\mu\gamma_p| \leq \alpha_m\beta_n + \beta_n\gamma_p$$

since

$$\alpha_m\beta_n \geq \max_{\mu < m} |\alpha_m\alpha_\mu - \alpha_\mu\beta_n|$$

and the relation $|\beta_n\alpha_\mu - \alpha_\mu\gamma_p| \leq \beta_n\gamma_p$ holds by the induction hypothesis ($\mu+n+p < m+n+p$, $\alpha_\mu\beta_n + \beta_n\gamma_p \geq \alpha_\mu\gamma_p$, $\alpha_\mu\gamma_p + \gamma_p\beta_n \geq \alpha_\mu\beta_n$).

Our matrices therefore form a *metric* space $\mathcal{U}$.

From (2), again by induction from $m+n < s$ to $m+n = s$, it follows that $\alpha_m\beta_n$ is a *continuous* function of the numbers a_{ik}, b_{ik} . For example, when $m=n$ and $b_{ik} \rightarrow a_{ik}$, $\alpha_n\beta_n$ converges to $\alpha_n\alpha_n = 0$. Thus every matrix α_n can be approximated arbitrarily closely by one with rational coefficients, which means that $\mathcal{U}$ is *separable*.

The space $\mathcal{U}$ is universal for all countable spaces, that is, it contains an isometric image of every countable space $\{p_1, p_2, \dots\}$. Indeed, putting $p_ip_k = a_{ik}$, the matrices

$$\alpha_n = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \cdots & \cdots & \cdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}$$

form a set $\{\alpha_1, \alpha_2, \dots\}$ isometric to that space. The space $\mathcal{V}$ obtained by *completion* of $\mathcal{U}$ is therefore universal for all separable spaces, that is, it contains an isometric image of every separable space. Like $\mathcal{U}$, the space $\mathcal{V}$ is separable ($\mathcal{U}$ is dense in $\mathcal{V}$).²

²A nonseparable universal space $\mathcal{W}$ for all separable spaces is much easier to construct. For example, take the space of all bounded sequences of numbers $\xi = (x_1, x_2, \dots)$ with distance $\xi\eta = \sup |x_n - y_n|$. If $\{p_1, p_2, \dots\}$ is a countable space and $p_ip_k = a_{ik}$, then the sequences $\xi_i = (a_{i1}, a_{i2}, \dots)$, which are not necessarily bounded at first, form a space $\{\xi_1, \xi_2, \dots\}$ isometric to it. Indeed, $\xi_i\xi_k = \sup_n |a_{in} - a_{kn}| \leq a_{ik}$, and the maximum is attained for $n=i$ or k . The bounded sequences $0, \xi_2 - \xi_1, \xi_3 - \xi_1, \dots$ also form such a space.

Incidentally, $\mathcal{U}$ itself is not complete, so that $\mathcal{U} \subset \mathcal{V}$. Consider, for example, all the matrices

$$\alpha_2 = \begin{pmatrix} 0 & a_{12} \\ a_{21} & 0 \end{pmatrix} = \begin{pmatrix} 0 & x \\ x & 0 \end{pmatrix} (x \geq 0);$$

Since $\alpha_2\beta_2 = |x - y|$, they form a set isometric to the half-line $x \geq 0$. The space $\mathcal{V}$ contains a set isometric to the whole line, and hence, for example, a point β whose distance from α_2 is $x + p$ (with p a positive constant) for every x . But this β is not an element of $\mathcal{U}$, that is, it is not a matrix β_n . For then, for $\nu = 2, \dots, n$, we would have

$$\beta_\nu\alpha_2 = \max|\beta_\nu\beta_1 - \beta_1\alpha_2|, |\beta_\nu\beta_2 - \beta_2\alpha_2|, \dots, |\beta_\nu\beta_{\nu-1} - \beta_{\nu-1}\alpha_2| (\beta_1\alpha_2 = x).$$

For sufficiently large x , for example $x \geq \beta_1\beta_2 + \beta_2\beta_3 + \dots + \beta_{n-1}\beta_n$, this implies $\beta_\nu\alpha_2 = x - \beta_1\beta_\nu$. Indeed, if this already holds for the values $1, 2, \dots, \nu - 1$, then

$$\beta_\nu\alpha_2 = \max[x - \beta_1\beta_\nu, x - \beta_1\beta_2 - \beta_2\beta_\nu, \dots, x - \beta_1\beta_{\nu-1} - \beta_{\nu-1}\beta_\nu] = x - \beta_1\beta_\nu,$$

Thus $\beta_n\alpha_2 = x - \beta_1\beta_n = x - p \neq x + p$.

HOMOGENEITY OF THE SPACE $\mathcal{V}$.

As before, we use the same Greek letter for matrices such that the one with the smaller index is a section of the other. When this is not assumed, we denote the matrices by Latin letters, and the index need not indicate the order of the matrix. For example,

$$a_1 = \alpha_n = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \cdots & \cdots & \cdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix},$$

A sequence $(\alpha_1, \alpha_2, \alpha_3, \dots)$ is called a *canonical* sequence. We denote isometry by $\sim$, so that

$$(a_1, \dots, a_2, \dots, a_n) \sim (b_1, b_2, \dots, b_n)$$

means $a_i a_k = b_i b_k$.

I. If $(\xi_1, \xi_2, \dots, \xi_{n-1}, \xi_n) \sim (\xi_1, \xi_2, \dots, \xi_{n-1}, \alpha_m)$ and all sections of α_m occur among the matrices $\xi_1, \dots, \xi_{n-1}$ (with equality understood in our sense), then $\xi_n = \alpha_m$. Indeed,

$$\xi_n\alpha_m = \max_{\nu < n, \mu < m} |\xi_n\xi_\nu - \xi_\nu\alpha_m|, |\xi_n\alpha_\mu - \alpha_\mu\alpha_m| = \max_{\nu < m} |\xi_n\xi_\nu - \xi_\nu\alpha_m| = 0.$$

In particular, from $(\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n, \gamma_1, \dots, \gamma_p, \dots) \sim (\xi_1, \xi_2, \dots)$ it follows inductively that each term on the left equals the corresponding term on the right. Every sequence $(\alpha_m, \beta_n, \gamma_p, \dots)$ is a subsequence of a canonical sequence.

II. If $(\xi_1, \dots, \xi_m, \xi_{m+1})$ is an initial segment of a canonical sequence and $(a_1, \dots, a_m) \sim (\xi_1, \dots, \xi_m)$, then there is a matrix a_{m+1} such that $(a_1, \dots, a_m, a_{m+1}) \sim (\xi_1, \dots, \xi_m, \xi_{m+1})$. The proof is sufficiently clear in the case $m = 3$. Starting with $(\alpha_m, \beta_n, \gamma_p) \sim (\xi_1, \xi_2, \xi_3)$, we must

construct a matrix δ_q satisfying $(\alpha_m, \beta_n, \gamma_p, \delta_q) \sim (\xi_1, \xi_2, \xi_3, \xi_4)$. We determine the matrices η and ζ so that

$$\begin{aligned} &(\alpha_m, \beta_n, \gamma_p, \alpha_1, \dots, \alpha_{m-1}, \beta_1, \dots, \beta_{n-1}, \gamma_1, \dots, \gamma_{p-1}) \sim \\ &(\eta_1, \eta_2, \eta_3, \eta_4, \dots, \eta_{m+2}, \eta_{m+3}, \dots, \eta_{m+n+1}, \eta_{m+n+2}, \dots, \eta_{m+n+p}) \\ &(\text{where } \eta_1 = \xi_1, \eta_2 = \xi_2, \eta_3 = \xi_3) \text{ and} \end{aligned}$$

$$\begin{aligned} &(\eta_4, \dots, \eta_{m+2}, \eta_1, \eta_{m+3}, \dots, \eta_{m+n+1}, \eta_2, \eta_{m+n+2}, \dots, \eta_{m+n+p}, \eta_3, \xi_4) \sim \\ &(\zeta_1, \dots, \zeta_{m-1}, \zeta_m, \dots, \zeta_{m+n-1}, \zeta_{m+n}, \dots, \zeta_{m+n+p-1}, \zeta_{m+n+p}, \zeta_{m+n+p+1}). \end{aligned}$$

Let $\zeta_{m+n+p+1} = \delta_q$. By permutation, we obtain

$$(\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n, \gamma_1, \dots, \gamma_p) \sim (\zeta_1, \dots, \zeta_{m+n+p}),$$

Thus, by I, these groups of points are equal, $\zeta_m = \alpha_m$, $\zeta_{m+n} = \beta_n$, $\zeta_{m+n+p} = \gamma_p$. We then have

$$\begin{aligned} (\alpha_m, \beta_n, \gamma_p, \delta_q) &= (\zeta_m, \zeta_{m+n}, \zeta_{m+n+p}, \zeta_{m+n+p+1}) \sim \\ &\sim (\eta_1, \eta_2, \eta_3, \xi_4) = (\xi_1, \xi_2, \xi_3, \xi_4). \end{aligned}$$

III. If $(a_1, \dots, a_m) \sim (b_1, \dots, b_m)$, then for every b_{m+1} there is an a_{m+1} satisfying $(a_1, \dots, a_m, a_{m+1}) \sim (b_1, \dots, b_m, b_{m+1})$. This follows from II by means of $(\xi_1, \dots, \xi_{m+1}) \sim (b_1, \dots, b_m, b_{m+1})$.

IV. If $(a_1, \dots, a_m) \sim (b_1, \dots, b_m)$, then there is an isometric mapping of $\mathcal{V}$ onto itself which sends a_1 to $b_1, \dots, a_m$ to b_m .³

Let $(a_1, \dots, a_m, a_{m+1}, \dots), (b_1, \dots, b_m, b_{m+1}, \dots)$ be two countable dense sets in $\mathcal{U}$. (For example, if $(c_1, c_2, \dots)$ is a dense set in $\mathcal{U}$, we can choose $a_{m+1} = b_{m+1} = c_1$, and so on.) Then determine $y_{m+1}, x_{m+1}, y_{m+2}, x_{m+2}, \dots$ successively so that

$$\begin{aligned} &(a_1, \dots, a_m, a_{m+1}, x_{m+1}, a_{m+2}, x_{m+2}, \dots) \sim \\ &(b_1, \dots, b_m, y_{m+1}, b_{m+1}, y_{m+2}, b_{m+2}, \dots) \end{aligned}$$

This produces two isometric dense sets in $\mathcal{V}$, which determine the mapping.

³[Here HAUSDORFF noted in the margin that IV still needs to be extended to the case where $a_1, \dots, a_m, b_1, \dots, b_m$ are points of $\mathcal{V}$.]